

4(a)(i)	label line to where white chromatid crosses over grey chromatid ;	1
4(a)(ii)	<i>any four from:</i> 1 (crossing over is) between non-sister chromatids ; 2 (non-sister chromatids have) different combinations of alleles ; 3 exchange alleles ; 4 (chromatids have) new combinations of alleles ; 5 linkage groups broken ; 6 <i>idea that</i> gametes have unique combination of alleles ; 7 random fusion of gametes (during fertilisation) ;	4
4(b)(i)	<i>any two from:</i> 1 (first) generation of offspring (from parental cross) ; 2 (parents are) homozygous / true bred / pure bred / pure breeding ; 3 F1 generation will be heterozygous ;	2

Question	Answer	Marks																																			
4(b)(ii)	<table><tr><th>phenotype</th><th>expected ratio</th><th>O</th><th>E</th><th>$O - E$</th><th>$(O - E)^2$</th><th>$\frac{(O - E)^2}{E}$</th></tr><tr><td>full band and rays</td><td>9</td><td>99</td><td>76.5</td><td>22.5</td><td>506.25</td><td>6.62</td></tr><tr><td>full band and no rays</td><td>3</td><td>21</td><td>25.5</td><td>-4.5</td><td>20.25</td><td>0.79</td></tr><tr><td>broken band and rays</td><td>3</td><td>8</td><td>25.5</td><td>-17.5</td><td>306.25</td><td>12.01</td></tr><tr><td>broken band and no rays</td><td>1</td><td>8</td><td>8.5</td><td>-0.5</td><td>0.25</td><td>0.03</td></tr></table> 1 mark for E and $O - E$ correct ; 1 mark for $(O - E)^2$ and $\frac{(O - E)^2}{E}$ correct ; $\chi^2 = 19.45$; ecf for 1 mistake in one box	phenotype	expected ratio	O	E	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$	full band and rays	9	99	76.5	22.5	506.25	6.62	full band and no rays	3	21	25.5	-4.5	20.25	0.79	broken band and rays	3	8	25.5	-17.5	306.25	12.01	broken band and no rays	1	8	8.5	-0.5	0.25	0.03	3
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4(b)(iii)	any three from: 19.45 / chi-squared value, is greater than, 7.81 / critical value (at $p = 0.05$) ; - results are / observed ratio is, <u>significantly</u> different to expected ratio ; - so differences are not due to chance ; - due to, increase in full band phenotypes / decrease in broken band phenotypes ;	3
4(b)(iv)	any two from: - epistasis ; - linkage ; - lethal allele ; - chance deviations ; - environmental effects ; - mutations ; - small sample size ;	2